

EEAIC - DCDC Converter Assignment 3

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I - Design Choices

1.a) Number of Cascaded Stages

The implementation of the Successive-Approximation Switched-Capacitor (SAR SC) converter is based on a resolution defined by $V_{IN}/2^N$, where N represents the number of stages. For the specified input voltage $V_{IN} = 2.4V$, a target granularity of 150mV is required. While an LSB-first (recursive) topology generally offers superior conduction loss performance, its resolution characteristics are not ideally suited for this specific application. For instance, an $N = 4$ LSB-first configuration would produce a coarse resolution, while an $N = 5$ configuration would lead to a significantly finer resolution that deviates from the target values, yielding output levels such as 0.075V (minimum) and 2.325V (maximum).

In contrast, the MSB-first approach allows the converter to remain centered around the midpoint of the conversion range and facilitates a precise 2 : 1 conversion ratio. Although this topology entails higher conduction losses due to the series-cascaded switch architecture, it enables the achievement of the desired 150mV granularity while utilizing fewer conversion units compared to the recursive alternative. Consequently, an MSB-first design with four cascaded stages was selected. This configuration provides 16 discrete voltage levels ($2^4 = 16$), which perfectly maps the 2.4V input to the required 150mV resolution steps.

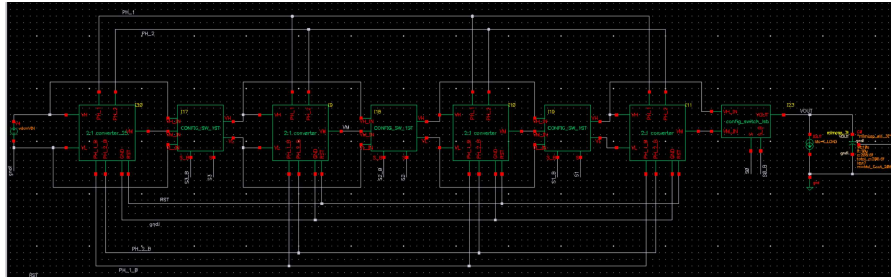


Figure 1: Schematic diagram of the implemented 4-stage MSB-first Successive-Approximation Switched-Capacitor (SAR SC) converter topology

1.b) Topology of each stage

Each individual stage of the converter is implemented using an interleaved, cross-coupled 2:1 Switched-Capacitor (SC) unit cell architecture. These cells are cascaded to recursively divide the input voltage, enabling high-resolution regulation with a significantly reduced component count compared to traditional ladder topologies. The stage topology operates as a three-terminal core bounded by an upper rail voltage (V_H), a lower rail voltage (V_L), and the regulated intermediate output voltage (V_M). Under ideal steady-state conditions, the target charge balance dictates the following voltage division relationship:

$$V_M = \frac{V_H + V_L}{2}$$

Structurally, each unit cell is split into two perfectly symmetrical, parallel conversion branches driven 180° out of phase to establish interleaved operation. During the first clock phase (Φ_1), the switching matrix configures the first branch to draw charge from the upper supply rail, while simultaneously forcing the second branch to discharge its stored energy directly into the intermediate output node V_M . During the second clock phase (Φ_2), the roles of the two branches are symmetrically reversed. This continuous current delivery mechanism ensures that the V_M node is actively driven across both clock phases, effectively doubling the output ripple frequency. To further enhance system stability and suppress high-frequency switching noise, a local decoupling capacitor is integrated between the intermediate V_M node of each stage and ground. This decoupling capacitor acts as a charge reservoir, damping dynamic transient spikes during dead-time transitions and keeping the intermediate rail voltage exceptionally stable to consistently meet the stringent 15 mV output ripple requirement.

The complete core layout of this unit cell is captured in Figure 2, while a detailed view highlighting the symmetrical internal transistor arrangement is provided in Figure 3. The main power transistors are driven through dedicated level shifters that integrate auxiliary reset switches. It is worth noting that the depicted schematic represents the standard stage topology, with an exception for the first stage where power transistors are implemented using 2.5 V-rated devices.

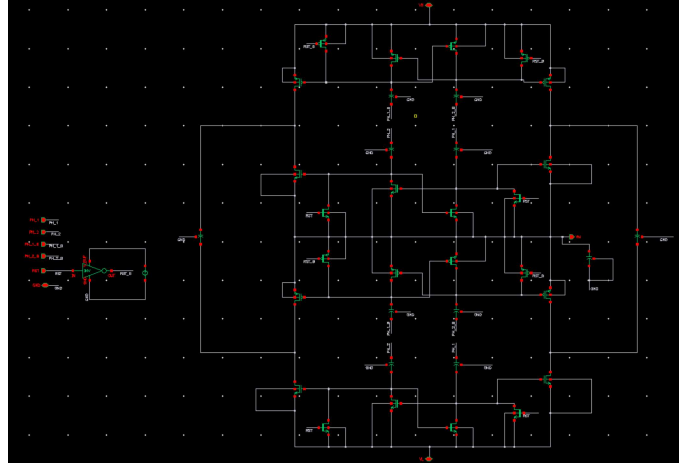


Figure 2: Complete schematic diagram of the interleaved, cross-coupled 2:1 Switched-Capacitor unit cell architecture

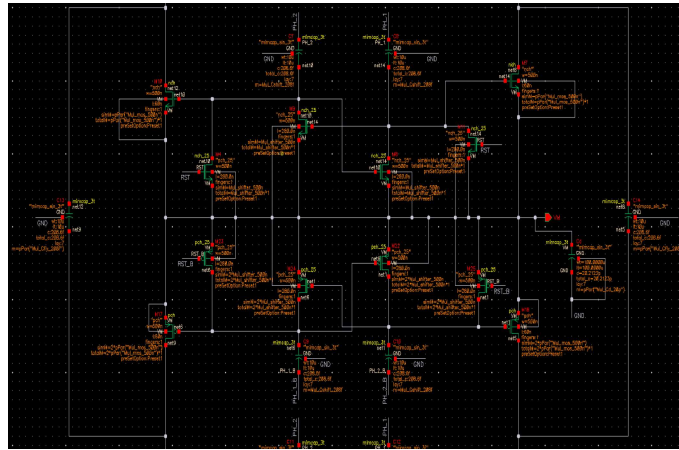


Figure 3: Detailed zoomed-in view of the 2:1 converter switching core showing the symmetrical branch device allocation

To build the complete multi-ratio converter, the individual stages are simply connected in series. Between each stage, configuration switch blocks route the signals dynamically, adjusting the input and reference voltages for the next cell in line. This allows the cascaded chain to step down the voltage precisely and hit the target output value.

For the intermediate stages, these configuration switches are implemented utilizing robust CMOS transmission gates, as illustrated in Figure 4. The parallel combination of NMOS and PMOS devices within these transmission gates ensures full rail-to-rail voltage pass-through capabilities with a low, uniform on-resistance R_{on} across all variable operating conditions.

An exception to this uniform implementation occurs at the final stage of the converter, which is associated with the Least Significant Bit (LSB). The final configuration block serves a distinct, specialized routing purpose: rather than feeding a subsequent cell, it must dynamically select whether to connect the global converter output node (V_{OUT}) to either the upper rail (V_H) or the intermediate split rail (V_M) of this final stage. Due to this reduced multiplexing requirement, a smaller routing topology is deployed for the LSB stage, the explicit schematic of which is detailed in Figure 5.

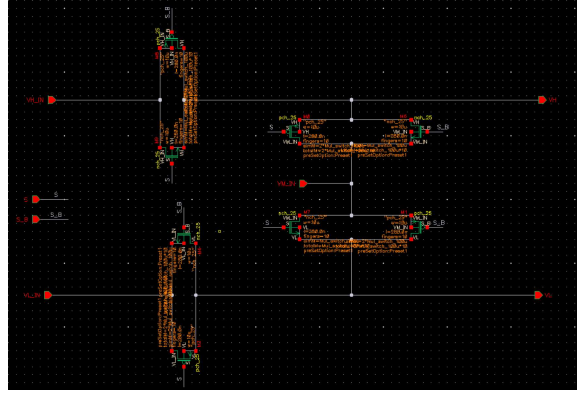


Figure 4: Schematic of the standard inter-stage configuration switch block implemented via CMOS transmission gates

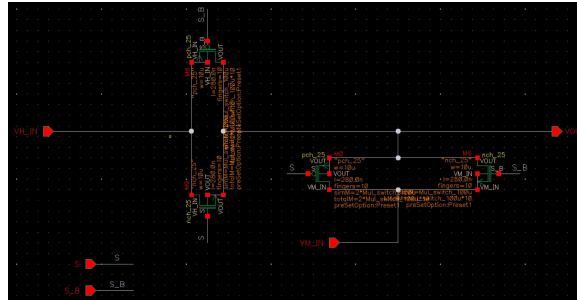


Figure 5: Schematic of the specialized LSB configuration switch block designed for multiplexing V_{OUT} between the final V_H and V_M nodes

1.c) Gate Driving Scheme and Level Shifting

The gate driving network is responsible for translating the ideal clock signals into appropriately referenced driving voltages for the power transistors across all cascaded stages. As specified in the assignment, the equal and opposite clock phases (Φ_1 and Φ_2) were generated using ideal `vpulse` voltage sources. To prevent shoot-through currents during

the switching transitions, a non-overlapping dead time equal to $1/100$ of the switching period (T_{sw}) was enforced between the two phases.

These ideal signals are initially passed through a custom-designed CMOS inverter to generate their negated counterparts ($\overline{\Phi_1}$ and $\overline{\Phi_2}$). A second identical inverter stage is cascaded to buffer the original clock phases. Implementing this dedicated buffer chain serves a dual purpose: it provides the necessary drive strength for the subsequent capacitive loads, and it establishes a distinct power domain (*e.g.*, a dedicated V_{BUF} supply). Isolating the buffers enables precise quantification of the switching losses (P_{sw}), which primarily stem from charging and discharging the level-shifter input capacitors and the power transistors' parasitic gate capacitances. Furthermore, operating this buffer chain at 1.2 V is critical for system efficiency; since dynamic switching losses scale quadratically with voltage ($P_{sw} \propto V^2$), utilizing a 2.4 V pulse would increment these logic switching losses by a factor of four.

Because the unit cells are cascaded and operate across varying floating potentials, the 1.2 V logic-level clock signals cannot directly drive the power MOSFETs. To address this, four capacitive level-shifters are embedded within each 2:1 converter stage. These blocks translate the ground-referenced logic phases to the specific local rail voltages (V_H , V_M , and V_L) of each transistor, ensuring safe and complete turn-on and turn-off transitions without exceeding the device oxide voltage limits.

Since the C_{shift} sole function is to transfer enough charge to toggle the internal cross-coupled latch, they were sized to be extremely small compared to both the flying capacitors (C_{fly}) and the decoupling capacitors (C_d). This minimizes their parasitic contribution to the strict 5 nF total capacitance budget while having a negligible impact on the global power transfer capability of the converter.

Finally, as illustrated in the schematic of Figure 3, a reset signal and associated circuitry were included in the level shifter design, as recommended in the course lecture notes. This ensures that the cross-coupled nodes initialize to a known, safe state during power-up, preventing latch-up and startup failures.

1.d) Device Types and Rating

Power Transistors

The device type and voltage rating for the main power transistors were determined by analyzing the maximum voltage differences experienced across their terminals throughout the entire operational timeline, accounting for both steady-state and transient phases.

While a steady-state analysis confirms that the terminal voltages of all power transistors naturally settle within a safe 1.2 V range, the first conversion stage is exposed to severe voltage stress during the initial startup phase. When the converter powers up with completely discharged internal capacitors ($t = 0$), the first-stage switches must briefly withstand the full input potential of 2.4 V. Consequently, robust 2.5 V devices (`nch.25` and `pch.25`) are utilized exclusively in Stage 1 to prevent destructive overvoltage oxide breakdown during these initial transients. The subsequent downstream stages, protected by the pre-divided voltage rails, safely exploit high-efficiency 1.2 V core devices

to minimize active switching and conduction losses.

Configuration Switches

The topology configuration switches are implemented entirely using CMOS transmission gates composed of 2.5 V devices, as illustrated in the schematic of Figure 4. Depending on the digital SAR control codes (0011, 0111, and 1011), these switches dynamically route intermediate potentials across the cascaded network.

When a specific configuration path is deactivated, the switch may be forced to isolate an open node sitting at the full 2.4 V supply rail while its gate control lines are held at the opposite absolute supply rail. Utilizing 1.2 V core devices in this scenario would induce severe, terminal overvoltage stress across both the gate oxide (V_{GS} , V_{GD}) and the drain-source channel (V_{DS}). To guarantee long-term oxide reliability and avoid channel punch-through under all blocked routing conditions, the 2.5 V device rating is mandatory for all configuration blocks.

Level Shifters

The design employs 2.5 V IO devices for the level-shifting circuitry to provide the necessary reliability against the 2.4 V input rail, preventing dielectric breakdown in the gate oxide. While this strict voltage stress constraint is primarily applicable to the first stage, these thicker-oxide IO devices are conservatively maintained across all subsequent downstream stages for design simplicity and universal layout reuse; it has been verified via simulation that this choice introduces no significant handicap and allows the system to successfully meet all the required performance specifications.

1.e) Allocation of the Total 5 nF Capacitance Budget

To systematically allocate the rigid 5 nF total capacitance constraint, a targeted sizing methodology is employed. The total capacitance budget is divided among four distinct capacitor classes: the main output filtering capacitance (C_{out}), the flying capacitors (C_{fly}), the intermediate decoupling capacitors (C_d), and the input coupling capacitors of the level-shifters (C_{shift}).

Main Output Filtering Capacitance (C_{out})

The primary objective for C_{out} is to suppress the output voltage fluctuations below the strict specification limit of $\Delta V_{ripple} \leq 15 \text{ mV}$ under a continuous current load of $I_{load} = 100 \mu\text{A}$.

Due to the interleaved 2:1 architecture, the output node is continuously fed across alternating clock phases, meaning the ripple frequency is effectively doubled ($2f_{sw}$). The first-order analytical relationship dictating the voltage drop during a single discharge cycle is given by:

$$\Delta V_{ripple} \approx \frac{I_{load} \cdot \Delta t}{C_{out}} = \frac{I_{load} \cdot T_{sw}}{2 \cdot C_{out}} = \frac{I_{load}}{2 \cdot f_{sw} \cdot C_{out}}$$

This equation highlights that the voltage ripple is directly proportional to the load current and the switching period T_{sw} and inversely proportional to the capacitance value. To satisfy safety margins and absorb high-frequency transient spikes, C_{out} must be allocated a substantial portion of the overall budget, target-sized in the 1.5 nF range.

- **Allocated Value:**
 $C_{out} = 1.435 \text{ nF}$
- **Percentage of Total Budget:** 28.7%

Flying Capacitances ($C_{fly,1...4}$)

The flying capacitors are allocated using a non-uniform strategy optimized specifically for the target conversion configurations (0.6 V, 1.2 V, and 1.8 V), which correspond to the SAR digital codes 0011, 0111, and 1011, respectively.

- **Stages 1 and 2 ($C_{fly,1}$ and $C_{fly,2}$):** Because these two early stages (MSBs) actively participate in charge pumping and voltage division across all targeted codes, they carry the primary current load. A key objective in sizing these components is to minimize the converter's slow-switching-limit resistance (R_{SSL}), which exhibits a strict inverse dependency on both switching frequency and capacitance ($R_{SSL} \propto 1/(f_{sw}C_{fly})$).

Furthermore, as noted previously, the current flowing through the second stage during operation in the 0.6 V and 1.8 V configurations is exactly double that of the first stage. To effectively balance this asymmetric current distribution and prevent excessive transient voltage sagging ($\Delta V = Q/C_{fly}$), $C_{fly,2}$ is intentionally sized to be twice the value of $C_{fly,1}$. This non-uniform distribution ensures that both active capacitances are scaled sufficiently high to mitigate conduction and redistribution losses.

- **Stages 3 and 4 ($C_{fly,3}$, $C_{fly,4}$):** For the specified target ratios, the LSB stages act purely as static pass-through channels. Since they do not undergo active charge redistribution, their flying capacitors handle negligible active switching current and are sized minimally to preserve the global capacitance budget.
- **Allocated Values:**
 $C_{fly,1} = 312 \text{ pF}$ — $C_{fly,2} = 624 \text{ pF}$
 $C_{fly,3} = 2.8 \text{ pF}$ — $C_{fly,4} = 2.8 \text{ pF}$
- **Total Flying Cap:** $2 \cdot 312 \text{ pF} + 2 \cdot 624 \text{ pF} + 4 \cdot 2.8 \text{ pF} = 1883.2 \text{ pF}$
- **Total Flying Cap Percentage of Budget:** 37.7%

Intermediate Buffer Capacitances ($C_{d,1...4}$)

The stage-by-stage buffer capacitors sustain the intermediate voltage rails (V_M) linking the cascaded blocks. Their sizing is directly governed by the charge-redistribution dynamics of each stage:

- **Stages 1 and 2 ($C_{d,1}, C_{d,2}$):** Sized higher to provide a sufficiently rigid charge reservoir for the active switching cycles of the MSB stages. Since these stages undergo heavy charge-pumping, larger capacitances are mandatory to minimize transient voltage drops (charge-redistribution ripple) on the intermediate rails, thereby preventing degradation of the overall converter efficiency and output resistance.
- **Stages 3 and 4 ($C_{d,3}, C_{d,4}$):** Sized lower. Stage 3 operates in pass-through mode, requiring minimal local charge storage. Similarly, $C_{d,4}$ can be kept minimal (scaled to match $C_{d,3}$) because the main output load is permanently connected to the V_H node of the final stage. Since the load bypasses the mid-voltage nodes (V_M) of the last two stages, the net output current sourced by these specific 2:1 SC cells is exactly zero. Consequently, these nodes experience no cyclic charge depletion, exhibiting negligible ripple and allowing us to safely minimize their capacitance to preserve the global area budget.

Design Summary:

- **Target Sizing Rule:** $C_{d,1}, C_{d,2} \gg C_{d,3}, C_{d,4}$
- **Allocated Values:**

$C_{d,1} = 800 \text{ pF}$	—	$C_{d,2} = 800 \text{ pF}$
$C_{d,3} = 20 \text{ pF}$	—	$C_{d,4} = 20 \text{ pF}$
- **Total Decoupling Cap** $1600 \text{ pF} + 40 \text{ pF} = 1640 \text{ pF}$
- **Total Decoupling Cap Percentage of Budget:** 32.8%

Level-Shifter Input Capacitances (C_{shift})

With 4 level-shifter blocks embedded per 2:1 cell and 2 differential phase capacitors per level shifter, there are 32 coupling capacitors in total across the 4-stage architecture.

Because the sole function of C_{shift} is to inject a small, instantaneous pulse of charge sufficient to toggle the internal cross-coupled latch of the level-shifter, their absolute values have virtually no impact on the power conversion capability of the converter. To prevent these 32 units from aggressively consuming the strict 5 nF global budget, they are sized minimally in the low femto-to-picoFarad range.

- **Allocated Value:** 1 pF per capacitor (Total for $32 \times C_{shift}$: 32 pF)
- **Percentage of Total Budget:** 0.6%

Capacitance Budget Summary

Once the exact values are finalized during the simulation and optimization phases, they can be tabulated in Table 1 to demonstrate strict compliance with the global 5 nF limit.

Table 1: Capacitance Budget Allocation Summary

Capacitor Class	Value [nF / pF]	Percentage of Budget [%]
Output Filter (C_{out})	1435 pF	28.7 %
Flying Caps ($C_{fly,1-4}$)	1883.2 pF	37.7 %
Decoupling Caps ($C_{d,1-4}$)	1640 pF	32.8 %
Level Shifters ($32 \times C_{shift}$)	32 pF	0.6 %
Total Allocated	4.990 nF	99.8 %

1.f) Transistor Sizing

Throughout the entire design, the PMOS transistors are sized with twice the width of their NMOS counterparts to balance the mobility discrepancy between holes and electrons, thereby ensuring symmetrical switching characteristics.

Power Transistors

To optimize the physical dimensions of these power switches, a targeted sizing strategy is executed based on the active current demands of the designated conversion configurations (0011, 0111, and 1011). Under these operating modes, Stage 1 and Stage 2 actively pump charge and deliver heavy current through their intermediate voltage outputs.

Charge multiplier analysis indicates that during operation in the 0.6 V and 1.8 V configurations, the current flowing through the second stage is exactly double that of the first stage. Consequently, a non-uniform scaling approach is adopted; the second stage is scaled up to feature larger transistors in order to minimize conduction losses ($I^2 R_{on}$) under this heightened current load, while still keeping dynamic switching losses and output voltage ripple sufficiently low.

Specifically, the power transistors in Stage 1, which utilize the thicker-oxide 2.5 V (.25) devices to withstand startup stress, are designed with a total effective width of 10 μm and a channel length of 280 nm. To accommodate the doubled current load, Stage 2 utilizes high-efficiency core devices configured with a significantly larger effective width of 20 μm and a shorter channel length of 60 nm.

Conversely, because the remaining downstream stages act purely as passive pass-through channels that do not actively split voltages or carry heavy switching currents under the target configurations, their power transistors are left at their small, default dimensions. This strategic downsizing minimizes the global parasitic gate area, keeping the clock distribution network's dynamic switching losses ($P_{sw} \propto f_{sw} C_g V^2$) strictly minimized where active charge transfer is absent.

Configuration Switches

Because the configuration switches sit directly in the primary power delivery paths between adjacent converter stages, they must carry the entire active load current transferred throughout the cascaded network. Any non-negligible series resistance within these connection paths would induce an unwanted voltage drop, directly degrading output voltage regulation and lowering the overall conversion efficiency. To mitigate this, the transmission gates are sized aggressively large to minimize their on-resistance and ensure an unattenuated voltage transfer between successive blocks.

Crucially, this large-scale sizing strategy introduces no penalty to the converter's dynamic power efficiency. These configuration switches operate semi-statically as they toggle exclusively when the target output voltage configuration is changed rather than at the high clock switching frequency f_{sw} . Therefore, their massive parasitic gate capacitances do not contribute to the dynamic switching loss overhead (P_{sw}), allowing the aspect ratios of the transmission gates to be maximized solely for minimum conduction resistance.

Level Shifters

The sizing of the level-shifting circuitry is governed by a critical trade-off between propagation delay and dynamic power consumption. These blocks must be sized sufficiently large to ensure sharp, fast rise and fall transitions at their outputs; optimizing the driving capability minimizes the rise and fall times of the subsequent power transistor gates, ensuring that the propagation delay does not consume the designated dead-time margin, which is critical to preventing shoot-through currents.

However, because the level shifters switch continuously at the clock frequency (f_{sw}), making them too large would introduce heavy parasitic capacitance and self-loading. To prevent this, we keep the internal transistors as small as possible to minimize their own dynamic switching losses. To meet these balanced performance requirements, the level-shifter devices are implemented with an effective width of 500 nm and a channel length of 280 nm, which is consistent with the mandatory minimum length for the thicker-oxide 2.5 V (_25) devices. This approach balances the design, providing enough drive strength for the power gates while keeping the auxiliary power consumption to a minimum.

1.g) Efficiency Evaluation and Frequency Optimization

The overall power efficiency (η) of the converter is calculated by factoring in both the primary conduction path and the overhead power dissipated by the driving circuitry. The efficiency percentage is defined as follows:

$$\eta = \frac{P_{out}}{P_{in} + P_{buf} + P_{buf,2}} \times 100$$

Where the individual power components are defined and extracted from the simulation environment as described below:

- **Output Power (P_{out}):** The average power delivered directly to the load.
- **Input Power (P_{in}):** The real power entering the system from the primary input voltage source (V_{IN}). To ensure steady-state accuracy, the instantaneous power is clipped and averaged over the last 20 switching cycles before the end of the simulation (T_{sim}). The exact Cadence Virtuoso Calculator expression utilized is:
`abs(average(clip((IT("/V4/PLUS") * VT("/VIN")) (VAR("Tsim") - (20 / VAR("fsw")))) VAR("Tsim"))))`
- **Dynamic Buffer Power (P_{buf}):** The power dissipated to dynamically charge and discharge the internal switch gate capacitances. This is evaluated by measuring the average power supplied by the voltage source dedicated to the signal inverters and gate buffers (V_{BUF1}). The corresponding simulation expression is:
`abs(average(clip((IT("/V0/PLUS") * VT("/V_BUF1")) (VAR("Tsim") - (20 / VAR("fsw")))) VAR("Tsim"))))`
- **Configuration Buffer Power ($P_{buf,2}$):** The power consumed by the buffers handling the static or low-frequency switch configuration signals. This contribution is calculated analogously to P_{buf} but is found to be negligible during standard steady-state operation.

The converter efficiency is heavily dependent on the switching frequency (f_{sw}), which introduces a well-known trade-off between switching losses (predominant at higher frequencies) and conduction/ripple losses (predominant at lower frequencies). To maximize the efficiency performance across all the required operating configurations, the switching frequency was optimized and set to $f_{sw} = 800$ kHz.

1.h) Design Achievements

The project successfully fulfilled all the design requirements. Each designated task was executed, meeting or exceeding the targeted criteria for validation.

II - Simulation Results

2.a) Transient Waveforms at Zero-Load Condition ($I_{LOAD} = 0$ A)

To verify the fundamental operational functionality and steady-state voltage balancing of the cascaded converter, the transient behavioral waveforms are analyzed under a zero-load condition ($I_{LOAD} = 0$ A). The digital SAR configuration switches are programmed to target a nominal output voltage of $V_{OUT} = 1.2$ V.

Figures 6, 7, and 8 display the comprehensive transient evolution of the system from startup ($t = 0$) until it fully settles into steady state. The plot captures the primary input rail ($V_{IN} = 2.4$ V), the regulated output node (V_{OUT}), and the internal intermediate node potentials across the cascaded cell structure, specifically tracking V_H , V_M , and V_L . Additionally, the two non-overlapping generated clock phases (Φ_1 and Φ_2) are superimposed to demonstrate proper charge-transfer synchronization.

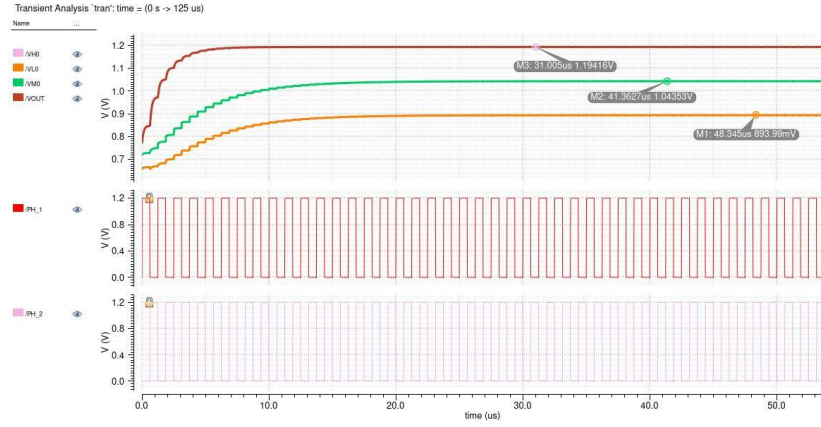


Figure 6: Transient simulation waveforms of V_{OUT} and the last converter's output V_{L0} , V_{M0} and V_{H0} , together with the generated phases

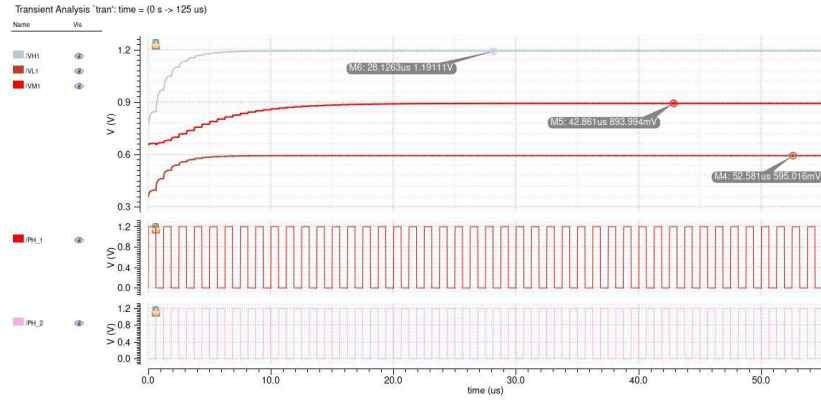


Figure 7: Transient simulation waveforms of V_{L1} , V_{M1} and V_{H1} , together with the generated phases

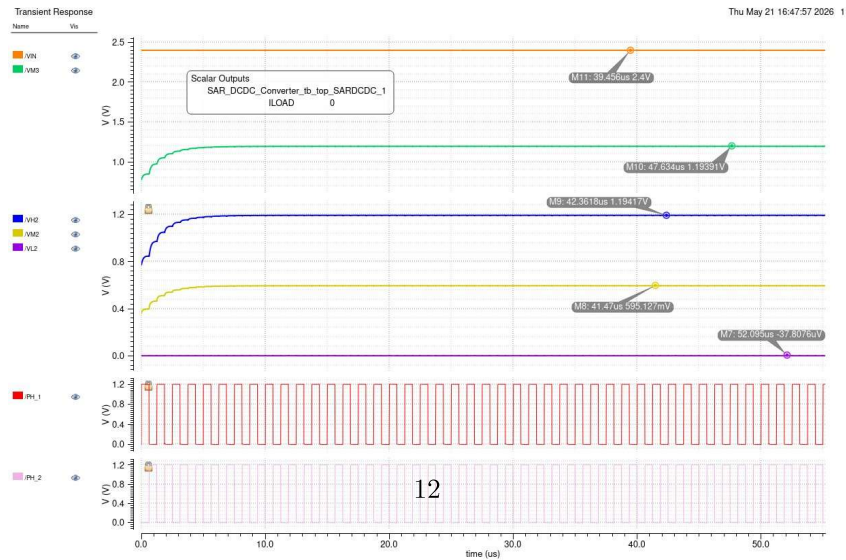


Figure 8: Transient simulation waveforms of V_{M3} and V_{IN} , V_{L2} , V_{M2} and V_{H2} , together with the generated phases

2.b) Configuration Sweep and Output Voltage Granularity

The reconfigurability of the SAR converter is evaluated by sweeping through the entire digital control code space to demonstrate the voltage regulation range. Simulations were performed at $I_{LOAD} = 0$ A to isolate the open-circuit voltage division ratios.

Figure 9 lists the average steady-state output voltages (V_{OUT}) extracted via the Cadence Virtuoso Calculator using a precise time-domain clip-and-average window. The results validate that the 4-stage cascaded architecture successfully achieves the targeted multi-ratio output profile, guaranteeing a maximum voltage steps granularity of 150 mV between successive configurations.

Point	Output	Nominal
Filter	Filter	Filter
Parameters: S3=0, S2=0, S1=0, S0=0		
1	VOUT_average_pss	147.6m
Parameters: S3=0, S2=0, S1=0, S0=1		
2	VOUT_average_pss	296.3m
Parameters: S3=0, S2=0, S1=1, S0=0		
3	VOUT_average_pss	445.6m
Parameters: S3=0, S2=0, S1=1, S0=1		
4	VOUT_average_pss	595.1m
Parameters: S3=0, S2=1, S1=0, S0=0		
5	VOUT_average_pss	743.8m
Parameters: S3=0, S2=1, S1=0, S0=1		
6	VOUT_average_pss	893.4m
Parameters: S3=0, S2=1, S1=1, S0=0		
7	VOUT_average_pss	1.044
Parameters: S3=0, S2=1, S1=1, S0=1		
8	VOUT_average_pss	1.194
Parameters: S3=1, S2=0, S1=0, S0=0		
9	VOUT_average_pss	1.347
Parameters: S3=1, S2=0, S1=0, S0=1		
10	VOUT_average_pss	1.496
Parameters: S3=1, S2=0, S1=1, S0=0		
11	VOUT_average_pss	1.647
Parameters: S3=1, S2=0, S1=1, S0=1		
12	VOUT_average_pss	1.797
Parameters: S3=1, S2=1, S1=0, S0=0		
13	VOUT_average_pss	1.947
Parameters: S3=1, S2=1, S1=0, S0=1		
14	VOUT_average_pss	2.097
Parameters: S3=1, S2=1, S1=1, S0=0		
15	VOUT_average_pss	2.249
Parameters: S3=1, S2=1, S1=1, S0=1		
16	VOUT_average_pss	2.4

Figure 9: Cadence Virtuoso Calculator output table displaying the extracted average steady-state V_{OUT} across all digital configuration codes at $I_{LOAD} = 0$ A

2.c) Efficiency Profile and Output Ripple

The efficiency profiles of the converter as a function of the load current, swept from 0 A to 100 μ A for the three required nominal output configurations ($V_{OUT} = 0.6$ V, 1.2 V, and 1.8 V), are presented below.

To accurately compute the overall system efficiency and properly quantify global losses, the total input power calculation accounts for both the main input rail supply and the dynamic power consumed by the inverters in the phase driving network. As observed in the sweeps, the efficiency increases alongside the load current. This trend occurs because the static contribution of the switching losses becomes progressively less dominant in the total power budget as the output conduction power scales up.

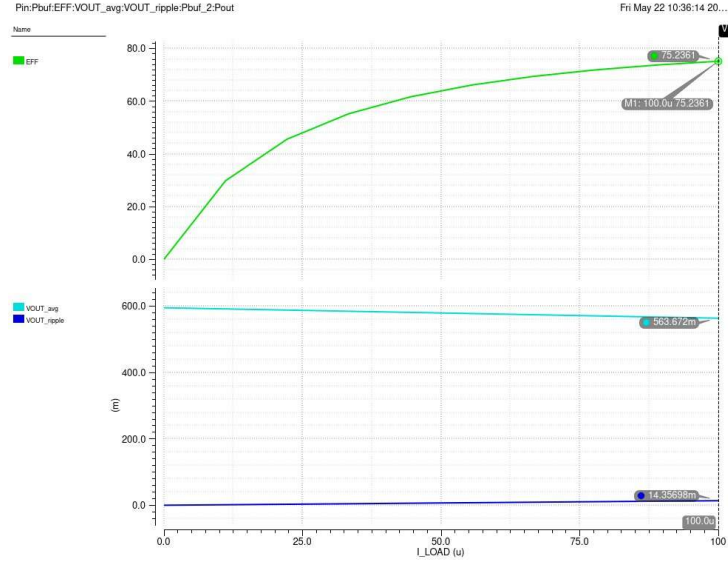


Figure 10: Efficiency profile as a function of I_{LOAD} for a nominal $V_{OUT} = 0.6$ V

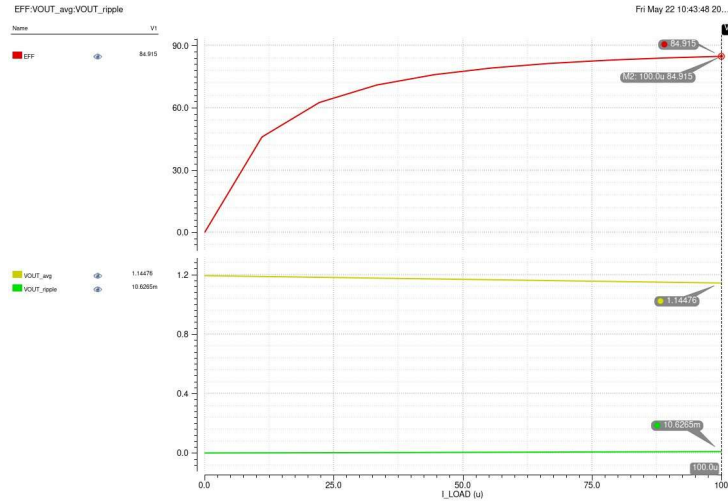


Figure 11: Efficiency profile as a function of I_{LOAD} for a nominal $V_{OUT} = 1.2$ V

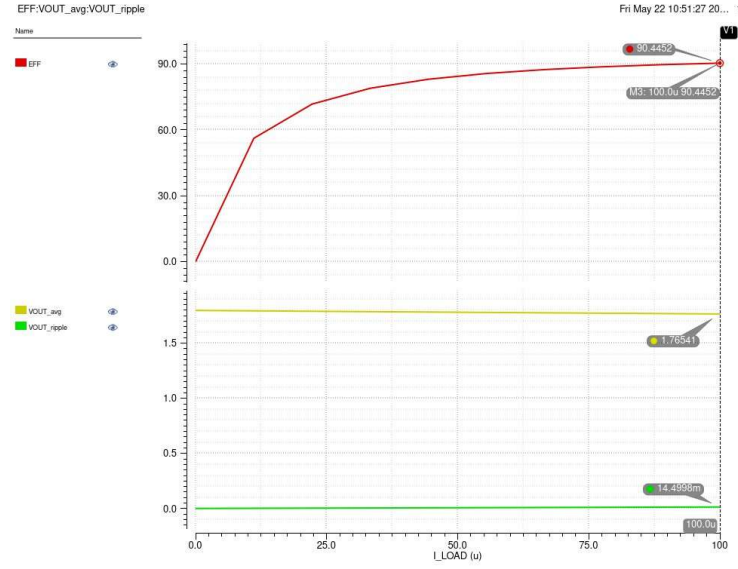


Figure 12: Efficiency profile as a function of I_{LOAD} for a nominal $V_{OUT} = 1.8$ V

The table below summarizes the extracted efficiency percentages along with their associated steady-state output voltage ripples at the maximum load condition, demonstrating full compliance with the required design specifications.

Point	Test	Output	Nominal
Filter	Filter	Filter	Filter
Parameters: S3=0, S2=0			
1	SAR_DCDC_Converter_tb_top_SARDCDC_1	EFF	75.24
1	SAR_DCDC_Converter_tb_top_SARDCDC_1	VOUT_avg	563.7m
1	SAR_DCDC_Converter_tb_top_SARDCDC_1	VOUT_ripple	14.36m
1	SAR_DCDC_Converter_tb_top_SARDCDC_1	/VOUT	
1	SAR_DCDC_Converter_tb_top_SARDCDC_1	/VIN	
Parameters: S3=0, S2=1			
2	SAR_DCDC_Converter_tb_top_SARDCDC_1	EFF	84.91
2	SAR_DCDC_Converter_tb_top_SARDCDC_1	VOUT_avg	1.145
2	SAR_DCDC_Converter_tb_top_SARDCDC_1	VOUT_ripple	10.63m
2	SAR_DCDC_Converter_tb_top_SARDCDC_1	/VOUT	
2	SAR_DCDC_Converter_tb_top_SARDCDC_1	/VIN	
Parameters: S3=1, S2=0			
3	SAR_DCDC_Converter_tb_top_SARDCDC_1	EFF	90.45
3	SAR_DCDC_Converter_tb_top_SARDCDC_1	VOUT_avg	1.765
3	SAR_DCDC_Converter_tb_top_SARDCDC_1	VOUT_ripple	14.5m
3	SAR_DCDC_Converter_tb_top_SARDCDC_1	/VOUT	
3	SAR_DCDC_Converter_tb_top_SARDCDC_1	/VIN	

Figure 13: Converter Efficiency and Output Voltage Ripple Performance at $I_{LOAD} = 100 \mu A$

III - Testbench setup

3.a) Testbench Plug and Play

TESTBENCH PATH : /home/eeaic_15fs26/assign6/dfii/SAR_DCDC_Converter/tb_top_SARDCDC

3.b) Verification Methodology of the Testbench

Verification of 2.a)

To observe the idealized transient behavior under zero-load conditions, the load current source must be set to $I_{\text{load}} = 0 \mu\text{A}$. To target an output voltage of $V_{\text{out}} = 1.2 \text{ V}$, the digital control bus must be driven with the binary word $S[3:0] = 0111_2$ ($S3 = 0$, $S2 = 1$, $S1 = 1$, $S0 = 1$). For full visualization of the switching behavior, the simulator's waveform outputs must be enabled for the following nodes:

/VOUT, /PH_1, /PH_2, /VIN, /VM3, /VH2, /VL2, /VM2, /VH1, /VL1, /VM1, /VH0, /VL0, /VM0

Verification of 2.b)

To map the complete Transfer Function and extract all 16 available output voltage levels, a parametric sweep is applied to the digital control word $S[3:0]$ from decimal 0 to 15. Within this setup, $S3$ is mapped as the Most Significant Bit (MSB) and $S0$ as the Least Significant Bit (LSB).

Verification of 2.c)

To evaluate system efficiency at the targeted voltage steps, the simulation must be executed across three distinct digital switch configurations corresponding to the desired output levels:

$V_{\text{out}} = 0.6 \text{ V}$: $S3 = 0$, $S2 = 0$

$V_{\text{out}} = 1.2 \text{ V}$: $S3 = 0$, $S2 = 1$

$V_{\text{out}} = 1.8 \text{ V}$: $S3 = 1$, $S2 = 0$

IMPORTANT NOTE: make sure to set `Mul_Cfly_208f`, `Mul_Cd_20p`, `Mul_mos_500n` as CDF variables for the 2:1 converter module and `Mul_Cfly_208f` and `Mul_Cd_20p` for the 2:1_converter_MSB as CDF variable from the Tool menu of the CIW, because we noticed that Cadence did not load them correctly every time we reopened the file, even if the last time were properly set.



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Energy-efficient Analog Integrated Circuits

DCDC Converters – Assignment 3

Assignment

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Introduction

In this assignment you will learn how to design a Successive-Approximation Switched-Capacitor DCDC converter topology (SAR), an attractive solution for several IoT applications where miniaturization and fine-grained conversion ratio resolution are required.

Please, refer to the following paper, as well as the course lecture notes:

S. Bang, D. Blaauw and D. Sylvester, "A Successive-Approximation Switched-Capacitor DC–DC Converter With Resolution of $V_{\text{IN}}/2^N$ for a Wide Range of Input and Output Voltages," in IEEE Journal of Solid-State Circuits, vol. 51, no. 2, pp. 543-556, Feb. 2016, doi: 10.1109/JSSC.2015.2501985.

The exercise will focus on the power stage and the gate drive schemes, while neglecting the phase generation and the control scheme for simplicity. You will not have to design the converter exactly as presented in the paper, please, follow the specifications listed below to choose the appropriate number of stages, stage topology, switch scheme, and so on.

Specifications of the Successive-Approximation Switched-Capacitor Step-Down DCDC Converter

I_{LOAD}=0

V _{IN}	2.4V
V _{OUT_nominal}	0.15V:0.15V:2.4V
V _{OUT_resolution_max}	150mV
C _{tot}	5nF

I_{LOAD}=100uA

V _{OUT_nominal}	EFF _{peak}	ΔV _{OUT}
0.6V	>75%	<15mV
1.2V	>80%	<15mV
1.8V	>85%	<15mV

QUESTIONS

- 1. In your report, explain the following features of your design on Cadence:**
 - a. Choice of the Number of cascaded stages
 - b. Topology of each stage
 - c. Gate driving scheme from the ideal vpulse to the power transistor gates
 - d. Choice of the transistor types and rating for the power transistors, configuration switches and level shifters
 - e. Allocation of the total 5nF capacitance
 - f. Sizing of the power transistors, configuration switches and level shifters
 - g. How the efficiency has been computed
 - h. Detail which specifications your design manages to accomplish and whether some are not fulfilled. In such case, please, explain how you proceeded.
- 2. Provide the following plots and data from your simulation results on Cadence:**
 - a. Transient waveforms of VIN, VOUT, each intermediate VH, VM and VL node, and both generated phases, when providing VOUT=1.2V with ILOAD=0A
 - b. A table printed from the cadence calculator of the average VOUT sweeping all the configurations when ILOAD=0A to show all the possible VOUT values and the desired VOUT maximum granularity of 150mV.
 - c. An efficiency plot of the three configurations providing a nominal VOUT of 0.6V, 1.2V and 1.8V, respectively, with ILOAD from 0 to 100uA, showing the efficiency profile and its maximum value in all three configurations. Please, add a printed table as well with the efficiency and the VOUT_ripple specifically at ILOAD=100uA.
- 3. Provide the path of your final testbench folder:**
 - a. Make sure that the provided testbench is plug and play, set on the configuration of point 2c, thus, providing the efficiency plots just by running and plotting the results.
 - b. Please, explain how you planned your testbench and what settings need to be changed to check the results as in the previous points 2a, 2b and 2c.